

Class 9 CBSE Information Technology (402)

Part B – Subject Specific Skills

Unit 1: Introduction to IT–ITeS Industry

Complete classroom notes • Simple English • Definitions, examples, tables, diagrams and revision questions



Learning objectives

After studying this unit, students will be able to:

- Explain the meaning of IT and ITeS.
- Identify different IT-enabled services and their uses.
- Understand outsourcing, BPO and BPM.
- Describe the structure of the IT–BPM industry.
- Explain the applications of IT in daily life and different fields.

1.1 Introduction

Information Technology (IT) is an important part of modern life. Computers, mobile phones, the internet and digital services help people work, study, communicate and solve problems quickly.

Today, many companies use computers and internet-based services to provide services to customers around the world. These services are known as IT-enabled services (ITeS).

What is the IT–ITeS industry?

The IT–ITeS industry is a sector that uses computers, software, communication networks and digital technologies to provide information technology products and services.

Examples of IT–ITeS industry



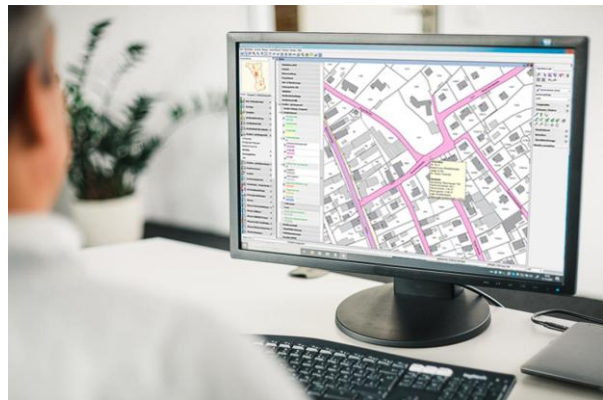
Software development



Customer support



Medical transcription



GIS mapping

1.2 What is IT?

Information Technology (IT) is the use of computers, software, networks and electronic devices to collect, store, process, retrieve and share information.

In simple words, IT helps us handle information using technology.

Main components of IT

Desktop computer system



1. Computer hardware

Physical devices such as computers, keyboards, monitors and printers.



2. Software

Programs and applications that perform tasks.



3. Computer networks

Systems that connect computers and allow data sharing.



4. Data and information

Facts and processed information stored and managed digitally.

Examples of IT in daily life

1. Using a computer to prepare a document.
2. Searching for information on the internet.
3. Sending an email.
4. Storing photos on a smartphone.
5. Using online banking.
6. Attending online classes.

Advantages of IT

- Saves time and effort.
- Improves accuracy.
- Makes communication faster.
- Stores large amounts of information.
- Helps automate repetitive tasks.
- Provides access to information from anywhere.

1.3 What is ITeS?

ITeS stands for Information Technology Enabled Services.

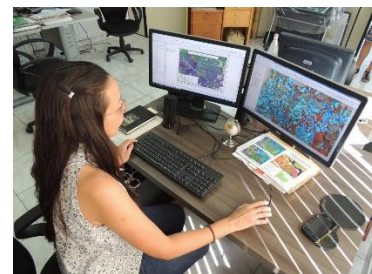
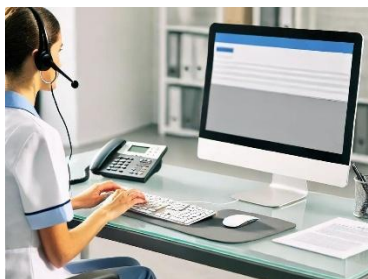
ITeS refers to services that use information technology, computers, software and communication networks to deliver services more efficiently.

In simple words, ITeS means using IT to provide services.

Difference between IT and ITeS

IT	ITeS
Information Technology	Information Technology Enabled Services
Focuses on technology, software, hardware and systems.	Focuses on services delivered with the help of IT.
Example: Developing a software application.	Example: Providing customer support using computers and communication networks.
Includes programming, networking and database management.	Includes call centers, medical transcription and digital document services.

Examples of ITeS



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- Call centers
- Medical transcription
- GIS mapping
- Enterprise Resource Planning (ERP)
- Knowledge management and archiving
- Online customer support

- Data processing
- E-publishing

1.4 Key IT-Enabled Services

1. Call centers

A call center is a service center where employees handle customer telephone calls using computers and communication systems.



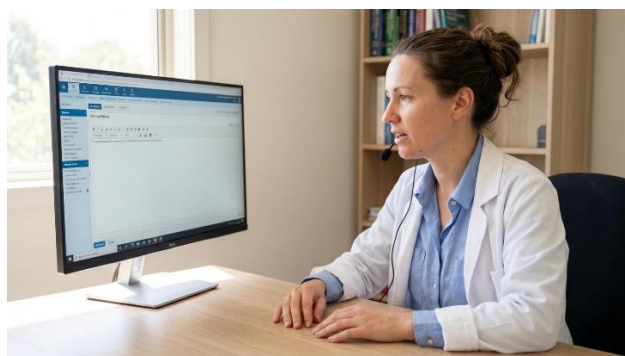
Services provided:

- Answering customer queries.
- Providing product information.
- Handling complaints.
- Technical support.
- Booking and appointment assistance.

Example: A customer calls a telecom company to ask about a recharge problem.

2. Medical transcription

Medical transcription is the process of converting recorded medical information, such as a doctor's dictation, into written electronic documents.



Uses:

- Preparing medical reports.
- Creating patient records.
- Maintaining hospital documentation.
- Converting voice recordings into text.

Example: A transcriptionist types a doctor's recorded diagnosis into a digital medical report.

3. GIS mapping

GIS stands for Geographic Information System.

GIS mapping is the use of computers and software to collect, store, analyze and display information related to locations on Earth.



Uses of GIS:

- Digital maps and navigation.
- City planning.
- Agriculture and land management.
- Disaster management.
- Transport planning.
- Environmental studies.

Example: Google Maps uses digital mapping technology to help people find locations and routes.

4. ERP (Enterprise Resource Planning)

ERP stands for Enterprise Resource Planning.

ERP is a software system that helps an organization manage and integrate its different business activities in one system.



ERP manages:

- Finance and accounting.
- Human resources.
- Inventory.
- Sales and purchasing.
- Production and operations.

Example: A company uses ERP software to track its stock, employee information and financial transactions.

5. Knowledge Management and Archiving

Knowledge management is the process of collecting, organizing, storing and sharing knowledge within an organization.

Archiving means storing important documents and information safely for future use.



Uses:

- Storing company documents.

- Organizing digital records.
- Sharing knowledge among employees.
- Preserving old files.
- Finding information quickly.

Example: A school stores its old examination records and notices in a digital archive.

Other IT-enabled services

Service	Meaning
Data entry	Entering information into a computer system.
Online customer support	Helping customers through websites, chat or email.
E-publishing	Publishing books, magazines and documents electronically.
Web promotion	Promoting websites, products or services online.
Software testing	Checking software for errors and defects.

Digital document management Creating, storing and managing electronic documents.

1.5 Outsourcing

Meaning of outsourcing

Outsourcing is the practice of hiring an outside company or service provider to perform a particular business activity or service instead of doing it within the organization.

In simple words, outsourcing means giving some work to another company that has the skills and resources to complete it.

Example of outsourcing

A company in the USA needs customer support. Instead of opening its own customer support office, it hires an Indian BPO company to handle customer calls.

How outsourcing works

Contracting company

Needs customer support services

Outsourcing agreement

Work is assigned to an outside provider

BPO company / Vendor

Provides customer support using IT

Customers receive support

Service is delivered on behalf of the contracting company

Important terms in outsourcing

1. Contracting company

A contracting company is the organization that hires an outside service provider to perform a specific task or business activity.

Example: A bank hires an external company to provide customer support.

2. BPO

BPO stands for Business Process Outsourcing.

BPO is the practice of outsourcing specific business processes to an external service provider.

Examples:

- Customer support.
- Data entry.
- Payroll processing.
- Accounting.
- Technical support.

3. BPO company / Vendor

A BPO company, also called a BPO vendor or service provider, is an organization that performs outsourced business processes for another company.

Example: A company that handles customer calls for a bank is a BPO vendor.

Benefits of outsourcing

1. Reduces operating costs.
2. Provides access to skilled workers.
3. Saves time.
4. Allows companies to focus on their main business.
5. Improves service availability.
6. Provides access to specialized technology.

1.5.1 Categories of ITeS Industry

The ITeS industry includes many types of services. Two important categories are:

ITeS Industry

Services delivered using information technology

BPO

Business Process Outsourcing

BPM

Business Process Management

A. BPO Industry

The BPO industry provides outsourced business services to organizations.

Major BPO services



1. Basic finance and accounting services

Bookkeeping, preparing invoices, recording expenses and maintaining financial records.



2. Taxation and insurance services

Processing tax-related records, insurance claims and policy information.



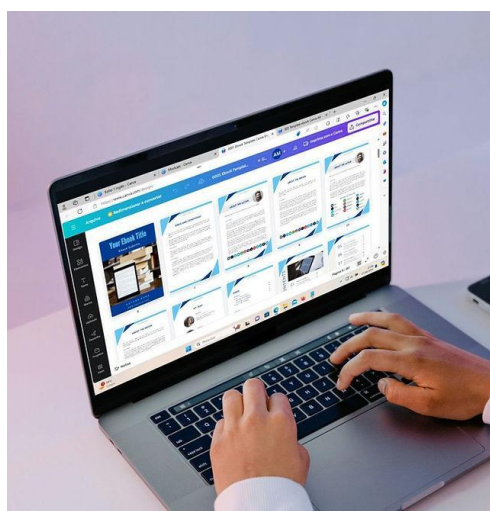
3. Legal services and content creation

Document preparation, legal research support and creating written digital content.



4. Basic human resource services

Maintaining employee records, payroll support and recruitment assistance.



5. E-publishing services

Preparing books, magazines and other documents for electronic publication.



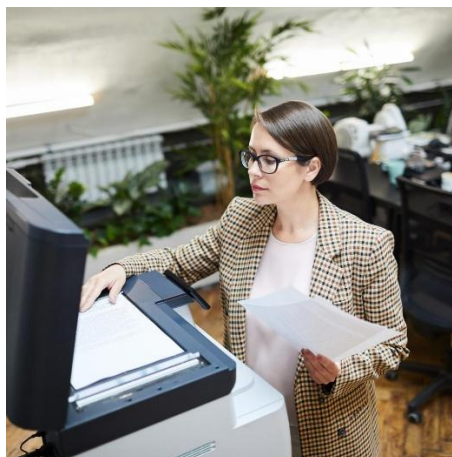
6. Web promotion services

Website promotion, digital marketing and online advertising support.



7. Multimedia and design services

Graphic design, video editing, animation and other creative digital services.



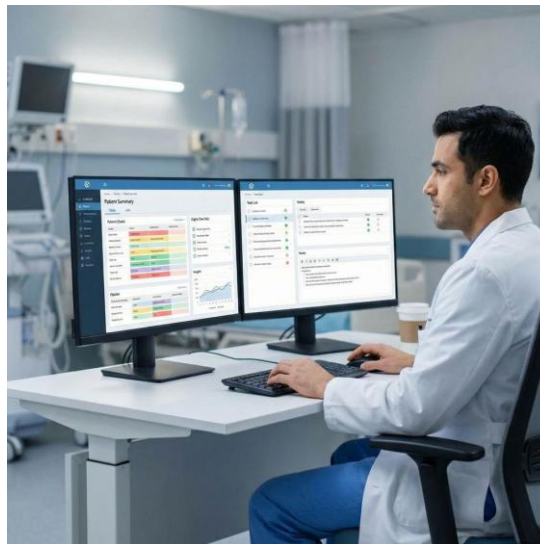
8. Document management services

Scanning, indexing, storing and retrieving digital documents.



9. Software testing services

Testing software applications to identify errors and improve quality.



10. Healthcare services

Medical billing, health records processing and healthcare administration support.

B. Call centers

A call center is an important part of the BPO industry.

Types of call centers:

Type	Meaning
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Inbound call center	Receives calls from customers.
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Outbound call center	Makes calls to customers.
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Inbound example: A customer calls a bank to ask about an ATM problem.

Outbound example: A company calls customers to inform them about a new product.

C. BPO Vendors

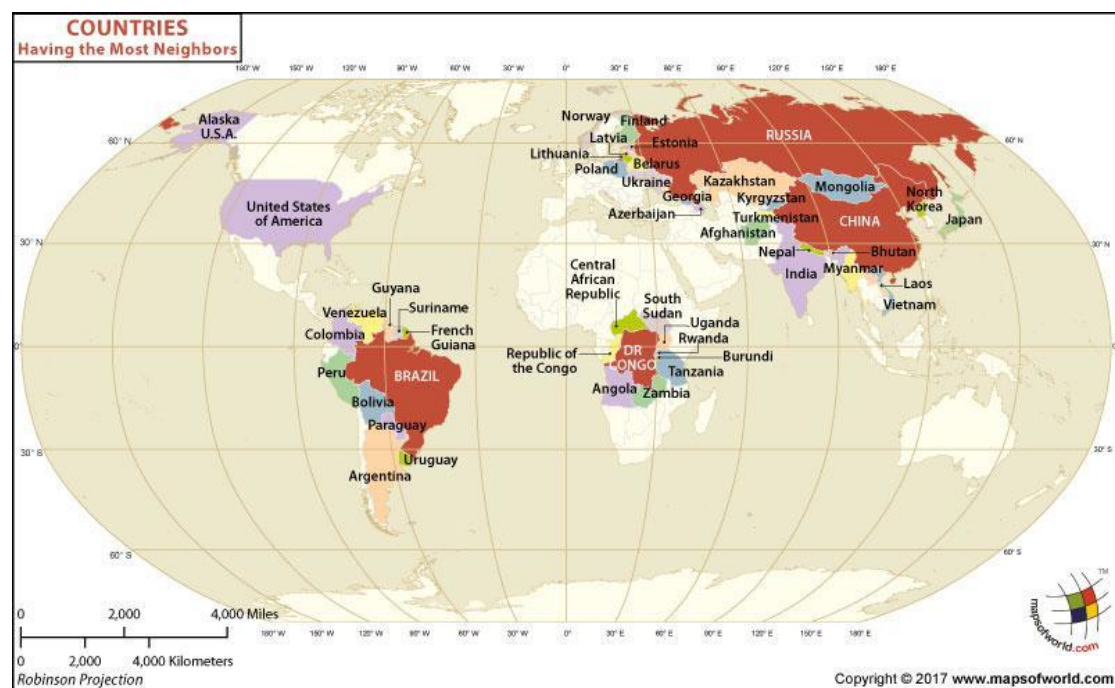
BPO vendors can be categorized according to their geographical location in relation to the contracting company.



1. Offshore vendor

An offshore vendor provides services from a different country.

Example: A US company hires an Indian BPO company.



2. Nearshore vendor

A nearshore vendor provides services from a nearby country or region.

Example: A company in the USA hires a BPO vendor in Mexico.



3. Onshore vendor

An onshore vendor provides services from the same country as the contracting company.

Example: A company in India hires a BPO vendor located in India.

Factors behind BPO industry success in India

India has become an important destination for BPO and IT services because of several factors.

1. Advanced technology

Availability of computers, high-speed internet, communication networks and modern software supports efficient service delivery.

2. Government support

Government initiatives, IT infrastructure development, skill development and policies supporting the IT sector have helped the industry grow.

3. Versatility

India has a large pool of skilled professionals who can provide services in different fields, languages and time zones.

Other supporting factors: English communication skills, a large workforce, educational institutions and the ability to provide services to international clients.

1.6 BPM Industry

What is BPM?

BPM stands for Business Process Management.

Business Process Management is the systematic approach of identifying, designing, executing, monitoring and improving business processes to make them more efficient.

In the IT–ITeS industry, BPM focuses on managing and improving business activities with the help of technology.

BPO vs BPM

BPO	BPM
Business Process Outsourcing.	Business Process Management.
Involves hiring an external company to perform business processes.	Involves managing, analyzing and improving business processes.
Focuses on providing outsourced services.	Focuses on improving the performance of processes.
Example: Hiring a vendor to process invoices.	Example: Analyzing and improving the invoice approval process.

Major areas of the BPM industry

1. Back-office transactions

Activities that support business operations but do not usually involve direct customer interaction.

- Data entry and record maintenance.
- Payroll processing.
- Finance and accounting transactions.
- Invoice processing.

2. IT and software operations

Technology-related services used to maintain and operate business systems.

- Software development support.
- Application maintenance.
- Technical support.
- Software testing.
- IT infrastructure monitoring.

3. Knowledge services

Services that require research, analysis and specialized knowledge.

- Market research.
- Data analysis.
- Research support.
- Business intelligence.

- Legal and financial research.

1.6.1 Structure of IT–BPM Industry

The IT–BPM industry can be categorized into three major groups:

IT–BPM Industry

MNCs

Multinational companies

ISPs

Indian service providers

GICs

Global in-house centers

1. MNCs – Multinational Companies

A Multinational Company (MNC) is a company that operates in more than one country.

In the IT–BPM industry, MNCs provide software, consulting, outsourcing and technology services across different countries.

Examples: IBM, Accenture, Microsoft.

2. ISP – Indian Service Providers

In this context, Indian Service Providers (ISPs) are Indian IT and IT-enabled service companies that provide technology and business services to clients.

Examples: Tata Consultancy Services, Infosys, Wipro.

Note: In networking, ISP commonly means Internet Service Provider. Here, ISP refers to Indian service providers in the IT–BPM industry.

3. GIC – Global In-house Centers

GIC stands for Global In-house Center.

A GIC is a center established and operated by a company to perform its own business or technology activities, often serving different countries.

Example: A multinational company establishes its own technology and accounting center in India to support its global operations.

Comparison

MNC	ISP	GIC
Operates across countries.	Indian IT service provider.	In-house center of a company.
Serves clients globally.	Provides services to clients.	Primarily supports its parent company.
Example: IBM.	Example: TCS.	Example: A company-owned global support center.

1.7 Applications of IT

Information Technology is used in almost every field of life. It helps people perform tasks faster, store information, communicate, analyze data and make decisions.

i. IT in home computing and everyday life



IT is used in homes for:

- Online shopping.
- Paying electricity and other bills.
- Online banking.
- Watching movies and listening to music.
- Online education.
- Video calls.
- Smart home devices.
- Booking tickets and travel.

Example: A person uses a smartphone to pay an electricity bill through an online payment application.

ii. IT in library automation and workplaces



Library automation

Library automation means using computers and software to manage library activities.

Applications:

1. Maintaining records of books.
2. Searching for books using a digital catalog.
3. Issuing and returning books.
4. Tracking overdue books.
5. Managing student and member records.
6. Barcode and RFID-based library systems.

IT in workplaces

- Preparing documents and presentations.
- Maintaining employee records.
- Online meetings.
- Managing attendance and payroll.
- Email communication.
- Data analysis and reporting.

Example: A school uses library management software to issue books and track returns.

iii. Applications in Business



IT helps businesses manage their activities and serve customers.

Applications:

- Online shopping and e-commerce.
- Inventory management.
- Accounting and billing.
- Customer relationship management.
- ERP systems.
- Business communication.
- Sales and marketing.
- Data analysis.

Example: A shop uses billing software to generate bills and maintain stock records.

iv. Medicine and Healthcare



IT plays an important role in healthcare.

Applications:

1. Electronic health records.
2. Telemedicine and online consultation.
3. Medical imaging and scanning.

4. Hospital management systems.
5. Medical research.
6. Online appointment booking.
7. Medical billing and reporting.

Example: A doctor uses a computer to view a patient's digital medical record.

v. Communications



IT has made communication fast and convenient.

Applications:

- Email.
- Instant messaging.
- Video conferencing.
- Social media.
- Voice and video calls.
- Online collaboration.
- Digital file sharing.

Example: Students attend an online class through a video conferencing application.

vi. Education and Reference



IT is widely used in education for teaching, learning and research.

Applications:

- Smart classrooms.
- Online classes.
- Digital textbooks and e-books.
- Educational websites.
- Online examinations.
- Digital libraries.
- Learning management systems.
- Reference materials and online encyclopedias.

Example: A student uses an educational website to learn Java programming.

vii. Science, Research and Engineering

IT helps scientists and engineers perform complex calculations, analyze data, create designs and make predictions.



1. Data analysis

Data analysis is the process of examining data to find useful information and patterns.

Example: Scientists analyze rainfall data to study climate patterns.

2. Prediction modelling

Prediction modelling uses data and mathematical or statistical techniques to estimate future outcomes.

Example: A weather model predicts the possibility of rainfall.

3. CAD – Computer-Aided Design

CAD stands for Computer-Aided Design.

CAD is the use of computers and software to create, modify and analyze designs.

Uses: Engineering drawings, buildings, machines and product designs.

4. CAE – Computer-Aided Engineering

CAE stands for Computer-Aided Engineering.

CAE uses computer software to analyze and simulate engineering designs and performance.

Example: Testing the strength of a bridge design using computer simulation.

5. CAM – Computer-Aided Manufacturing

CAM stands for Computer-Aided Manufacturing.

CAM uses computers and software to control or support manufacturing processes.

Example: Computer-controlled machines producing machine parts in a factory.

6. Complex mathematical calculations

Computers perform large and complicated calculations quickly.

Example: Scientists use computers to solve mathematical equations and analyze scientific data.

7. Cosmos and astronomy studies

IT is used to study space, planets, stars and galaxies.

Applications:

- Analyzing telescope data.
- Studying planets and stars.
- Space mission planning.
- Simulating the movement of celestial objects.
- Processing satellite images.

Example: Astronomers use computers to analyze images of distant galaxies.

viii. Law Enforcement



Law enforcement agencies use IT to maintain records, investigate crimes and improve public safety.

Applications:

- Crime records management.
- CCTV surveillance.
- Digital forensics.
- Cybercrime investigation.
- Fingerprint and biometric systems.
- Traffic monitoring.
- Emergency response systems.

Example: Police use digital databases to search crime records and identify suspects.

ix. Government and Public Service



IT is used by governments to provide services to citizens.

Applications:

- Online government portals.
- Digital certificates.

- Online tax payment.
- Public grievance systems.
- Digital identity services.
- E-governance.
- Online application forms.
- Land and property records.

Example: A citizen applies for a government certificate through an online portal.

x. Banking and Insurance



IT in banking

- ATM services.
- Internet banking.
- Mobile banking.
- Online fund transfers.
- Digital account management.
- Banking transaction records.
- Fraud monitoring.

IT in insurance

- Online policy purchase.
- Policy management.
- Insurance claim processing.
- Customer records.
- Premium payment.
- Risk analysis.

Example: A customer transfers money using mobile banking without visiting a bank.

xi. Military



IT is used in military organizations for communication, planning, training and information management.

Applications:

- Secure communication.
- Military training simulations.
- Radar and surveillance systems.
- Navigation.
- Satellite communication.
- Logistics and supply management.
- Cybersecurity.
- Data analysis.

Example: A military organization uses computer systems to monitor and coordinate communication.

xii. IT Applications in Marketing



IT has changed the way businesses promote products and services.

Applications:

1. Digital advertising.
2. Social media marketing.
3. Email marketing.
4. Search engine optimization (SEO).
5. Online surveys.
6. Customer data analysis.
7. E-commerce marketing.
8. Website promotion.

Example: A business promotes its products through social media advertisements.

xiii. Sports



IT is used in sports to improve performance, analyze matches and support decisions.

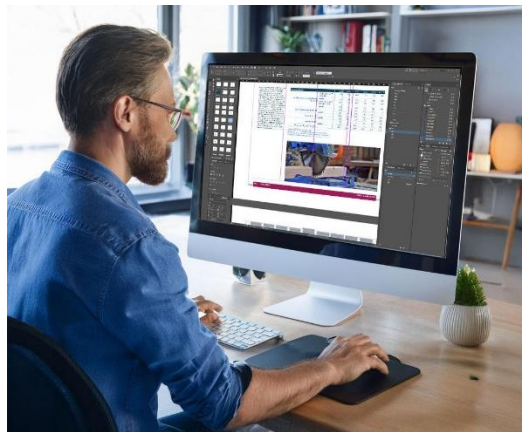
Applications:

- Player performance analysis.
- Scoreboards and score management.

- Video replay.
- Sports statistics.
- Fitness tracking.
- Match scheduling.
- Digital ticket booking.
- Decision-support technologies.

Example: Cricket teams use video analysis and computer statistics to study player performance.

xiv. Publishing and Printing



IT is used to create, edit, publish and print books, newspapers, magazines and other documents.

Applications:

- Desktop publishing.
- Book and magazine layout.
- Digital printing.
- E-books.
- Online newspapers.
- Photo editing.
- Document formatting.
- Typesetting.

Example: A publisher uses desktop publishing software to design a textbook before printing it.

xv. Music, Arts, Dance and Entertainment



IT is widely used in creative and entertainment industries.

Music

- Recording and editing songs.
- Digital audio production.
- Music streaming.
- Creating beats and sound effects.
- Online music distribution.

Arts

- Digital painting.

- Graphic design.
- Animation.
- Photo editing.
- 3D modelling.

Dance

- Music and choreography preparation.
- Digital video recording.
- Online dance classes.
- Stage lighting and sound control.

Entertainment

- Film editing.
- Computer-generated imagery (CGI).
- Animation.
- Video games.
- Online streaming.
- Special effects.

Example: A music producer uses digital audio software to record and edit a song.

Important differences for examination

1. IT vs ITeS

IT	ITeS
Technology used to manage information.	Services delivered using IT.
Example: Software development.	Example: Call center service.

2. BPO vs BPM

BPO	BPM
Business Process Outsourcing.	Business Process Management.
Outsourcing business work to another company.	Managing and improving business processes.
Example: Outsourced payroll processing.	Example: Improving the payroll process.

3. Offshore vs Nearshore vs Onshore

Offshore	Nearshore	Onshore
Different country.	Nearby country or region.	Same country.

4. CAD vs CAE vs CAM

Term	Full form	Main purpose
CAD	Computer-Aided Design	Creating and modifying designs.
CAE	Computer-Aided Engineering	Analyzing and simulating designs.
CAM	Computer-Aided Manufacturing	Supporting computer-controlled manufacturing.

Key terms to remember

Abbreviation Full form

IT	Information Technology
ITeS	Information Technology Enabled Services
BPO	Business Process Outsourcing
BPM	Business Process Management
ERP	Enterprise Resource Planning
GIS	Geographic Information System
MNC	Multinational Company
ISP	Indian Service Provider
GIC	Global In-house Center
CAD	Computer-Aided Design
CAE	Computer-Aided Engineering
CAM	Computer-Aided Manufacturing
SEO	Search Engine Optimization

Chapter summary

- IT is the use of computers, software and networks to manage information.
- ITeS means services provided using information technology.

- Outsourcing is hiring an outside company to perform a business activity.
- BPO means Business Process Outsourcing.
- BPM means Business Process Management.
- Call centers, medical transcription and GIS mapping are examples of ITeS.
- BPO vendors may be offshore, nearshore or onshore.
- The IT–BPM industry includes MNCs, Indian service providers and GICs.
- IT is used in homes, schools, business, healthcare, banking, science, government, sports and entertainment.

Exercise: Test your knowledge

MCQ Practice

Question 1 of 10

What is the full form of IT?

Information Technology

Internet Transfer

Information Transmission

Integrated Technology

PreviousNext

Short-answer questions

1. What is Information Technology?
2. Define ITeS with two examples.
3. What is outsourcing?
4. Write the full forms of BPO and BPM.
5. Explain medical transcription.
6. What is GIS mapping?
7. What is ERP?
8. Write three benefits of outsourcing.
9. Differentiate between onshore and offshore vendors.
10. Write three factors behind the success of the BPO industry in India.

11. Explain the three major areas of the BPM industry.
12. What is an MNC?
13. What is a Global In-house Center?
14. Write five applications of IT in education.
15. Explain the uses of IT in banking and insurance.

Teacher's note: Suggested screenshots for the student's notebook

For a polished CBSE classroom handout, insert screenshots of actual software interfaces under these topics:

Topic	Suggested screenshot
IT	Computer desktop or file management screen
Call centers	Customer support software dashboard
Medical transcription	Electronic medical record interface
GIS	Digital map with marked locations
ERP	Inventory or accounting dashboard
Library automation	Library management software
CAD	Engineering drawing in CAD software
CAM	Computer-controlled manufacturing machine
E-learning	Online classroom screen
Digital banking	Online banking interface (use a sample/demo screen, not personal banking details)

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